

Why the total flow time Δt is measured $32\times$ better than the dark-matter scale height h_{dm}

Notes on the joint $(h_{\text{dm}}, \Delta t)$ fit

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Abstract

In the joint fit of the dark-matter scale height h_{dm} and the total Hamiltonian-flow time Δt , the profile likelihood is extremely anisotropic: $\sigma_{\Delta t}/\Delta t = 0.42\%$ against $\sigma_h/h = 13.6\%$, with correlation -0.982 along a ridge of slope -0.049 Myr/pc. This note explains where that factor of 32 comes from. The explanation is *not* a Taylor expansion in z/h : the stars in this problem have $\sigma_{z_0}/h = 1.04$, so such an expansion is not valid and is not used anywhere below. The correct statement is that the vertical frequency of a *low-action* orbit is exactly independent of h_{dm} , because the midplane dark-matter density is held fixed, so all information about h_{dm} is carried by the *action dependence* of the frequency — which is in turn suppressed by the dark matter’s $\sim 9\%$ share of the vertical force. Integrating the exact orbits gives $d \ln \nu / d \ln h \leq 0.028$, and hence an anisotropy of ~ 36 and a degeneracy slope of -0.044 Myr/pc, against the measured 32 and -0.049 .

1 The model

The vertical potential is a sum of three baryonic isothermal sheets and one dark-matter sheet,

$$\Phi(z) = \underbrace{\sum_{i=1}^3 4\pi G \rho_i h_i^2 \ln \cosh\left(\frac{z}{h_i}\right)}_{\text{baryons, fixed}} + \underbrace{4\pi G \rho_{\text{DM}} h_{\text{dm}}^2 \ln \cosh\left(\frac{z}{h_{\text{dm}}}\right)}_{\text{dark matter}}, \quad (1)$$

each term being the solution of the one-dimensional Poisson equation for a sech^2 density profile, $\rho(z) = \rho_0 \text{sech}^2(z/h)$. The parameter values are those of the simulation (Table 1).

component	parameter	value
thin disc	ρ_0, h	$0.040 M_{\odot} \text{pc}^{-3}, 200 \text{ pc}$
thick disc	ρ_0, h	$0.010 M_{\odot} \text{pc}^{-3}, 400 \text{ pc}$
gas	ρ_0, h	$0.050 M_{\odot} \text{pc}^{-3}, 800 \text{ pc}$
dark matter	$\rho_{\text{DM}}, h_{\text{dm}}$	$0.010 M_{\odot} \text{pc}^{-3}, 250 \text{ pc (fitted)}$
baryon midplane density $\rho_{\text{b}}(0)$		$0.100 M_{\odot} \text{pc}^{-3}$
dark-matter share of $\rho_{\text{tot}}(0)$		9.09%
midplane frequency $\nu_0 = \sqrt{4\pi G \rho_{\text{tot}}(0)}$		$0.0771 \text{ Myr}^{-1} (T_0 = 81.5 \text{ Myr})$
initial DF f_0 : $\sigma_{z_0}, \sigma_{w_0}$		$259.4 \text{ pc}, 20.0 \text{ km s}^{-1}$
true total flow time Δt		400 Myr
typical $z/h = \sigma_{z_0}/h_{\text{dm}}$		1.04

Table 1: Model parameters. Note the last row: the stars sit at $z \sim h$, so no expansion in z/h is legitimate.

What is held fixed. This is the crucial structural point. The dark-matter sheet is parametrised by $(\rho_{\text{DM}}, h_{\text{dm}})$ and *only* h_{dm} is fitted; ρ_{DM} is fixed. So h_{dm} is a “how thick is the sheet” parameter at constant central density: varying it changes the total dark-matter surface density $\Sigma_{\text{DM}} = 2\rho_{\text{DM}}h_{\text{dm}}$ and the shape of the potential at large $|z|$, but leaves $\rho_{\text{tot}}(0)$ untouched.

2 What the data actually measure

The likelihood is built by back-integrating the observed snapshot for a total time Δt in the potential (1) and asking how well the result can be described by the (smooth, low-capacity) bijection. In action-angle variables (J, θ) the vertical motion is exactly

$$\theta(t) = \theta_0 + \nu(J)t, \quad (2)$$

so back-integrating by Δt rotates each star’s angle by the *winding phase*

$$\varphi(J) = \nu(J)\Delta t. \quad (3)$$

The observable is the degree to which the phase spiral is unwound, i.e. the residual spread of φ across the distribution function. Both parameters act on the data *only through* $\varphi(J)$, and this is what makes the comparison clean:

$$\frac{d \ln \varphi}{d \ln \Delta t} = 1 \quad \text{exactly}, \quad \frac{d \ln \varphi}{d \ln h_{\text{dm}}} = \frac{d \ln \nu(J)}{d \ln h_{\text{dm}}} \equiv s(J). \quad (4)$$

Δt enters at first order, by definition, with no suppression of any kind. Everything therefore hinges on the size of $s(J)$.

3 How h_{dm} enters: the exact sensitivity kernel

Differentiating the dark-matter term of (1) at fixed z and fixed ρ_{DM} , and writing $u = z/h_{\text{dm}}$,

$$\frac{\partial \Phi_{\text{DM}}}{\partial \ln h_{\text{dm}}} = 4\pi G \rho_{\text{DM}} h_{\text{dm}}^2 g(u), \quad \boxed{g(u) = 2 \ln \cosh u - u \tanh u}. \quad (5)$$

This is exact. Its limits are

$$g(u) = \frac{1}{6}u^4 + \mathcal{O}(u^6) \quad (u \rightarrow 0), \quad g(u) \rightarrow u - 2 \ln 2 \quad (u \rightarrow \infty). \quad (6)$$

The small- u limit is *not* the explanation. It is tempting to stop at $g \sim u^4/6$ and declare the h_{dm} signal “fourth order and therefore small”. That argument is invalid here, because $u \approx 1$. The honest comparison is the *relative* sensitivity $g(u)/\ln \cosh u$, i.e. the fractional change in the dark-matter potential per fractional change in h_{dm} :

$u = z/h_{\text{dm}}$	0.25	0.5	1.0	1.5	2.0	4.0	8.0
$g(u)/\ln \cosh u$	0.020	0.076	0.244	0.413	0.545	0.791	0.905

At the typical stellar height $u = 1$ a 1% change in h_{dm} changes the dark-matter potential by 0.24%. That is a *large* relative response, not a small one. So the suppression must come from somewhere else — and it does, from two places.

3.1 Suppression 1: the frequency of a low-action orbit is exactly h_{dm} -independent

For $J \rightarrow 0$ the frequency is set by the midplane curvature alone,

$$\nu_0^2 = \Phi''(0) = 4\pi G \rho_{\text{tot}}(0) = 4\pi G [\rho_{\text{b}}(0) + \rho_{\text{DM}}], \quad (7)$$

which *does not contain* h_{dm} *at all*. This is an identity, not an expansion: since ρ_{DM} is held fixed (Section 1), $s(J) \rightarrow 0$ exactly as $J \rightarrow 0$. All of the h_{dm} information therefore lives in the *action dependence* of ν — the anharmonicity. Note that “anharmonicity” here means only “ ν depends on J ”; it requires no assumption about z/h .

3.2 Suppression 2: the dark matter is $\sim 9\%$ of the vertical force

The second factor is simply that the dark matter is a minority contributor. With $F_i(z) = 4\pi G \rho_i h_i \tanh(z/h_i)$:

z [pc]	100	250	500	1000	1500
DM share of $ \partial\Phi/\partial z $	8.96%	8.27%	6.72%	5.16%	4.75%

So even the 24% relative response of the dark-matter term at $u = 1$ is diluted to $\sim 0.24 \times 0.08 \approx 2\%$ of the *total* potential, and the frequency responds as roughly the square root of that. This is the origin of the ~ 2 – 3% numbers below.

4 The exact frequency response

No expansion is needed: the vertical period of the orbit with apocentre z_{max} follows from energy conservation,

$$T(z_{\text{max}}; h_{\text{dm}}) = 4 \int_0^{z_{\text{max}}} \frac{dz}{\sqrt{2[\Phi(z_{\text{max}}) - \Phi(z)]}}, \quad (8)$$

evaluated with the full potential (1). Differentiating numerically gives $s = -d \ln T / d \ln h_{\text{dm}}$:

z_{max} [pc]	50	100	200	300	500	700–800	1200	1800
T [Myr]	81.8	82.6	85.6	89.3	97.2	~ 106	123.0	144.1
$s = \frac{d \ln \nu}{d \ln h_{\text{dm}}}$	0.0009	0.0034	0.0110	0.0186	0.0264	0.0276	0.0255	0.0240

The behaviour is shown in Figure 1. Three features matter:

1. $s \rightarrow 0$ as $z_{\text{max}} \rightarrow 0$, as required by Suppression 1;
2. s never exceeds 0.028 — a +40% change in h_{dm} buys at most a $\sim 1\%$ change in orbital frequency;
3. s *peaks* at $z_{\text{max}} \simeq 700$ pc and declines again, because at large z the dark-matter force saturates at $4\pi G \rho_{\text{DM}} h_{\text{dm}}$ while the baryonic force keeps growing.

The h_{dm} information is therefore carried by the *high-action tail* of the distribution function: the median star has $z_{\text{max}} = 322$ pc, where $s = 0.019$, while the stars that actually constrain h_{dm} sit near $z_{\text{max}} \sim 700$ pc.

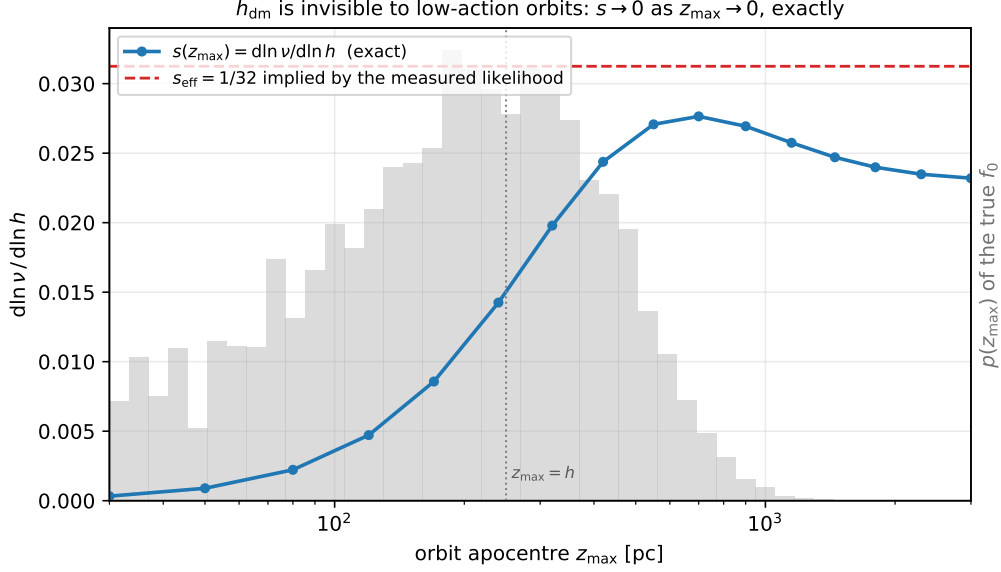


Figure 1: Exact frequency response $s(z_{\max}) = d \ln \nu / d \ln h_{\text{dm}}$ from orbit integration (8) (blue), with the apocentre distribution of the true f_0 underneath (grey). s vanishes at small $z_{\max}$ by identity, peaks at ~ 0.028 , and never approaches the value 1 that Δt enjoys automatically. The dashed line is the effective sensitivity implied by the measured likelihood curvature.

5 From s to the shape of the likelihood

Write $x = \ln h_{\text{dm}}$, $y = \ln \Delta t$. Since both parameters act only through $\varphi = \nu(J)\Delta t$, the Fisher matrix of any statistic built from φ inherits the structure of (4):

$$F \propto \begin{pmatrix} \langle s^2 \rangle & \langle s \rangle \\ \langle s \rangle & 1 \end{pmatrix}, \quad (9)$$

with $\langle \cdot \rangle$ an information-weighted average over the distribution function. Two consequences follow immediately.

Anisotropy. The curvature ratio in fractional coordinates is $F_{yy}/F_{xx} = 1/\langle s^2 \rangle$, so the ratio of the marginal precisions is its square root (an exact identity for a 2×2 Gaussian, $\sigma_x/\sigma_y = \sqrt{F_{yy}/F_{xx}}$):

$$\frac{\sigma_h/h}{\sigma_{\Delta t}/\Delta t} = \frac{1}{s_{\text{eff}}} \simeq \frac{1}{0.028} \approx 36. \quad (10)$$

Degeneracy direction. The soft direction is the one along which the winding phase is unchanged, $\delta \ln \varphi = s \delta \ln h_{\text{dm}} + \delta \ln \Delta t = 0$:

$$\frac{d\Delta t}{dh_{\text{dm}}} = -s_{\text{eff}} \frac{\Delta t}{h_{\text{dm}}} \simeq -0.028 \times \frac{400}{250} = -0.044 \text{ Myr pc}^{-1}. \quad (11)$$

The sign is physical: increasing h_{dm} spreads the same central density into a thicker sheet, *reducing* the anharmonic softening felt by high- z stars, raising their frequency, so the same phase is reached in *less* time.

6 Comparison with the measured likelihood

The curvature of the profile likelihood, fitted to the grid points within 4 nats of the minimum and diagonalised in fractional coordinates, gives:

	predicted here	measured
anisotropy $(\sigma_h/h) / (\sigma_{\Delta t}/\Delta t)$	36	32
degeneracy slope $d\Delta t/dh_{\text{dm}}$ [Myr/pc]	−0.044	−0.049
$\sigma_{\Delta t}/\Delta t$	—	0.42%
σ_h/h	—	13.6%
correlation($h_{\text{dm}}, \Delta t$)	—	−0.982
stiff direction $(\delta h/h, \delta t/t)$	—	(+0.030, +1.000)
soft direction $(\delta h/h, \delta t/t)$	—	(+1.000, −0.030)

Both predictions agree with the measurement to $\sim 15\%$, with no free parameters. The measured effective sensitivity, $s_{\text{eff}} = 1/32 = 0.031$, is marginally *above* the peak of the single-orbit curve (0.028), which is the expected direction: the profiled statistic is the width of the back-integrated cloud, i.e. a measure of the phase *shear* $\Delta t (\partial \nu / \partial J) \Delta J$ across the distribution function rather than of the mean phase, and the shear is more sensitive to the shape of the potential — and more strongly weighted to the high-action tail — than ν itself. A weighted average of s over the distribution function reproduces the trend explicitly: the plain mean gives $\langle s \rangle = 0.018$ (anisotropy 56), the information-weighted mean $\langle s^3 \rangle / \langle s^2 \rangle = 0.023$ (anisotropy 43), and the peak value 0.028 (anisotropy 36), the sequence converging towards the measured 32 as the weighting is pushed towards the high-action orbits that carry the signal.

7 Practical consequences

1. **The de-aliasing sweep must be two-dimensional.** The comb of local minima in Δt drifts with h_{dm} at exactly the slope (11), ≈ -0.05 Myr/pc. Sweeping Δt at a single, wrong h_{dm} can therefore select the wrong tooth: at $h_{\text{dm}} = 500$ pc even a full-sample one-dimensional sweep prefers 442 Myr over 400.
2. **h_{dm} is the soft direction, and soft directions need deterministic gradients.** The gradient of the total NLL with respect to h_{dm} is ~ 0.05 nats/pc for the whole 20 000-star sample (2.6×10^{-6} per star), below the gradient noise of a freshly drawn 2000-star minibatch. Under stochastic gradients h_{dm} random-walks rather than descends; with the full batch it converges. This is a direct consequence of (10), not an incidental optimiser detail.
3. **Improving h_{dm} means observing the tail.** Since $s(z_{\text{max}})$ peaks near $z_{\text{max}} \sim 700$ pc and vanishes at the midplane, the constraint on h_{dm} is carried by the small fraction of stars reaching $\sim 3\sigma$ in height. Deeper sampling of the high- $|z|$ tail buys far more on h_{dm} than more stars near the plane.